

## Part A · Multiple Choice

2 points each

1. **a) move under its own weight**

p. 275 — the text specifies at least tens of meters thick and hundreds of meters in extent, lasting decades or more. About 10% of Earth's land is currently glaciated; at times in the past million years it was at least 30%.

2. **c) About 69%**

p. 275 — the largest repository of freshwater on the planet, and highly sensitive to a warming climate.

3. **d) the “Snowball Earth” glaciations**

p. 277 — it is hypothesized the entire planet froze, with ocean ice up to 1 km thick even at the equator. The two main pulses were the Sturtian (~700 Ma) and the Marinoan (~650 Ma).

4. **a) The opening of the Drake Passage between South America and Antarctica**

p. 279 — unrestricted west-to-east flow isolated the Southern Ocean from warmer Pacific, Atlantic, and Indian water. Glaciers had begun forming on Antarctica by 35 Ma.

5. **b) the equilibrium line**

p. 281 — above it, not all the winter snow melts each summer; below it, more ice melts than accumulates. Its position shifts from year to year.

6. **b) 50 to 100 m**

p. 283 — ice deforms plastically at stresses around 100 kilopascals. The upper 50–100 m stays rigid, which is why it cracks into crevasses where the plastic ice below changes shape.

## Part B · True or False

1 point each

7. **TRUE**

p. 282 — both more winter snow and less summer melting favor advance, but of the two variables it is the summer melt that matters most.

8. **FALSE**

Ice is relatively soft — Mohs hardness 1.5 at 0°C. The erosion is done by rock fragments embedded in the ice and pressed against the bedrock, like sandpaper rather than plain paper. (p. 285)

## Part C · Short Answer

3 points each

9. **Above the equilibrium line, snow that does not melt is buried and compacted by later snowfall. It compresses into firn, in which snowflakes lose their shape and become granules; further compression squeezes the air out and welds the granules into glacial ice.**

**Rubric:** 1 pt — above the equilibrium line not all of the winter snow melts, so each year's layer is buried by the next and gradually compressed. 1 pt — the snow first becomes *firn*, within which the delicate snowflakes lose their shape and become granules. 1 pt — continued compression pushes the granules together, squeezes out the trapped air, and welds them into glacial ice. Credit the note that meltwater percolating down from the surface contributes to the process. (p. 281)

**10. U-shaped valleys, arêtes, cols, horns, cirques, hanging valleys, and truncated spurs.**

**Rubric:** 0.75 pt each for any four, requiring name and description. **U-shaped valley** — wide, flat-bottomed, steep-sided valley, since glaciers erode more at the base than the sides; the key feature of alpine glacial erosion. **Arête** — a sharp ridge between two glacial valleys. **Col** — a low point along an arête forming a pass between valleys. **Horn** — a steep peak eroded on three or more sides. **Cirque** — a bowl-shaped basin at the head of a glacial valley. **Hanging valley** — a tributary glacier's valley left above the main valley, which was eroded more deeply. **Truncated spur** — the end of an arête cut into a steep triangular cliff. Also accept drumlins, roches moutonnées, and glacial grooves or striations. (pp. 286–287)